# Session 3: Data Structures

Trainer Reference Guide

## Purpose and outcomes

Choose lists, tuples, sets, or dictionaries based on ordering, mutability, uniqueness, and key-based access; work with nested records and comprehensions.

## Examples and exercises

- 01_lists_tuples_sets.ipynb practices indexing, slicing, append, immutable coordinates, and unique departments.
- 02_dictionaries.ipynb creates and updates a user, loops through items, safely reads optional keys, and filters nested data.
- 03_nested_data.py searches users by ID and derives active names/departments.
- exercise_01_collections.py is guided modification, slicing, deduplication, and tuple work.
- exercise_02_user_search.py independently finds a matching dictionary or reports absence.
- challenge_inventory.py combines CRUD-like inventory operations and a low-stock comprehension.
- contact_manager_cli.py is Project 3: add, list, search, and delete contact dictionaries through a menu loop.

## Trainer key points specific to this module

Ask learners to justify the data structure before coding. Contrast an index with a dictionary key. Demonstrate ‘.get()‘ for optional values and warn against mutating a list during iteration. Trace one contact from input to nested dictionary to displayed result.

## Assessment evidence

The contact manager handles empty collections, case-insensitive search, missing deletes, and repeated actions.

## Delivery pattern

1. Explain the idea briefly and ask learners to predict a result.
2. Run the simple example, then connect it to an employee/user workflow.
3. Pair learners for the guided task; require an individual attempt next.
4. Use the challenge as the cumulative project increment.
5. Review the quiz and ask learners to explain, not only show, their code.

## Trainer key points

- Keep live coding visible and deliberate. Normalize reading tracebacks.
- Ask what type and value each expression produces before running it.
- Let TODO files remain student-owned; open solutions only during review.
- Check boundary and failure cases, not only the happy path.
- End by tracing how the topic appears in the final portal.